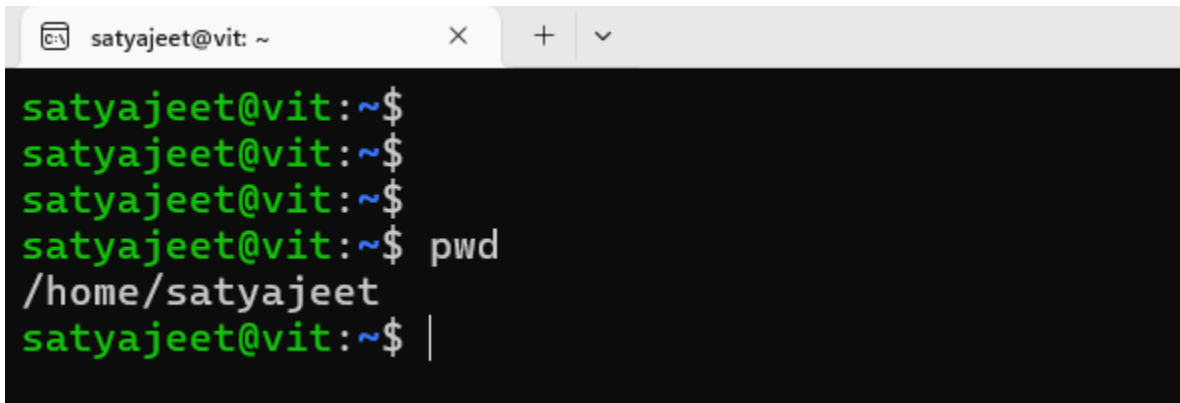
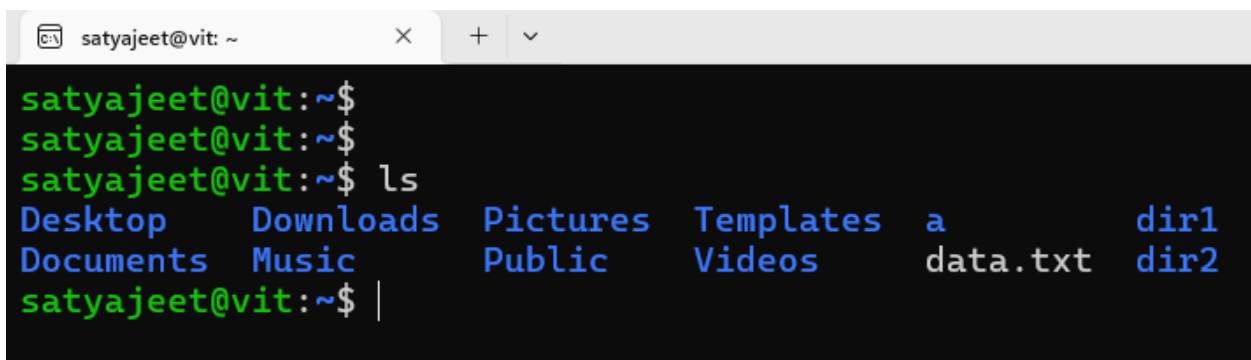


Linux Basics: Essential Commands, File Operations, and Permissions



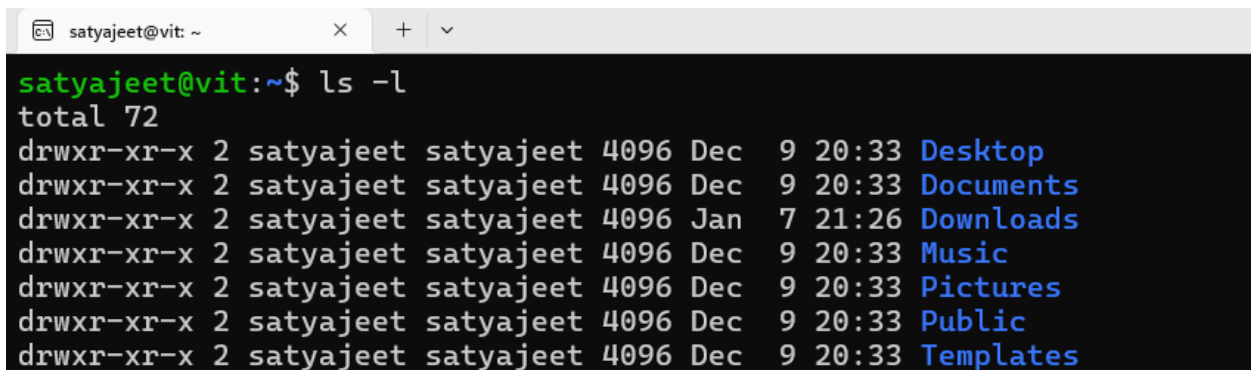
```
satyajeet@vit: ~  
satyajeet@vit:~$  
satyajeet@vit:~$  
satyajeet@vit:~$ pwd  
/home/satyajeet  
satyajeet@vit:~$ |
```

1. **pwd** – Displays the absolute path of the current working directory.



```
satyajeet@vit: ~  
satyajeet@vit:~$  
satyajeet@vit:~$ ls  
Desktop    Downloads  Pictures   Templates  a          dir1  
Documents  Music      Public     Videos    data.txt   dir2  
satyajeet@vit:~$ |
```

2. **ls** – Lists the files and directories present in the current directory.



```
satyajeet@vit: ~  
satyajeet@vit:~$ ls -l  
total 72  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Desktop  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Documents  
drwxr-xr-x 2 satyajeet satyajeet 4096 Jan  7 21:26 Downloads  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Music  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Pictures  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Public  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec  9 20:33 Templates
```

3. **ls -l** – Displays files and directories in long listing format with detailed information such as permissions, ownership, size, and timestamps.

```
satyajeet@vit: ~  
satyajeet@vit:~$ ls -a  
.          .bashrc    .local  
..         .cache     .motd_shown  
.ICEauthority .config    .profile  
.Xauthority  .dotnet    .sudo_as_admin_successful  
.bash_history .gnupg     .viminfo  
.bash_logout .lessht    .vscode-server  
satyajeet@vit:~$ |
```

4. **ls -a** – Lists all files and directories in the current directory, including hidden files (those starting with a dot).

```
satyajeet@vit: ~/Documents  
satyajeet@vit:~$  
satyajeet@vit:~$  
satyajeet@vit:~$  
satyajeet@vit:~$ cd Documents/  
satyajeet@vit:~/Documents$ |
```

5. **cd folder_name** – Changes the current working directory to the specified directory.

```
satyajeet@vit: ~  
satyajeet@vit:~/Documents$  
satyajeet@vit:~/Documents$  
satyajeet@vit:~/Documents$  
satyajeet@vit:~/Documents$ cd ..  
satyajeet@vit:~$ |
```

6. **cd ..** – Moves one level up to the parent directory.

```
satyajeet@vit: ~  
satyajeet@vit:~$  
satyajeet@vit:~$ mkdir docs  
satyajeet@vit:~$ ls  
Desktop    Downloads  Pictures   Templates  a          dir1  docs  
Documents  Music      Public     Videos    data.txt  dir2  file2  
satyajeet@vit:~$ |
```

7. **mkdir folder_name** – Creates a new directory with the specified name in the current location.

```
satyajeet@vit: ~/a/b  
satyajeet@vit:~$ mkdir -p a/b/c  
satyajeet@vit:~$ ls  
Desktop    Downloads  Pictures   Templates  a          data.txt  
Documents  Music      Public     Videos  
satyajeet@vit:~$ cd a  
satyajeet@vit:~/a$ ls  
b  
satyajeet@vit:~/a$ cd b  
satyajeet@vit:~/a/b$ ls  
c  
satyajeet@vit:~/a/b$ |
```

8. **mkdir -p folder_name/subfolder_name/subfolder_name** – Creates a directory structure with parent and child directories in a single command. The **-p** option in the **mkdir** command creates parent directories automatically if they do not already exist.

It also prevents errors if the specified directories already exist, allowing the entire directory path to be created in one step.

```
satyajeet@vit: ~  
satyajeet@vit:~$ touch file.txt  
satyajeet@vit:~$ ls  
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  a  file.txt  
thinclient_drives  
satyajeet@vit:~$ cat file.txt  
satyajeet@vit:~$ |
```

9. touch file_name – Creates a new empty file with the specified name if it does not already exist. As shown, running the cat command displays no output because the file does not contain any data.

```
satyajeet@vit: ~$ ls
Desktop  Downloads  Pictures  Templates  a      file1  thinclient_drives
Documents Music      Public    Videos    file.txt test

satyajeet@vit:~$ cp file1 file2
satyajeet@vit:~$ ls
Desktop  Downloads  Pictures  Templates  a      file1  test
Documents Music      Public    Videos    file.txt file2  thinclient_drives
satyajeet@vit:~$ |
```

10. cp source_file destination_file – Copies the contents of one file into another file.

As shown here, The contents of the source **file1** was copied into a new file **file2**. When the destination file is displayed using the cat command, it shows the same text, confirming that the copy operation was successful.

```
satyajeet@vit: ~/dir2$ ls
Desktop  Downloads  Pictures  Templates  a      file.txt  file2
Documents Music      Public    Videos    dir1    file1     test

satyajeet@vit:~$ cp -r dir1 dir2
satyajeet@vit:~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file1
Documents Music      Public    Videos    dir1    file.txt  file2

satyajeet@vit:~$ cd dir2
satyajeet@vit:~/dir2$ ls
data.txt  info.txt  java
satyajeet@vit:~/dir2$ |
```

11. cp -r dir1 dir2 – Copies a directory and all its subdirectories and files recursively to another directory.

As shown here, the source directory **dir1** was copied into the destination directory **dir2** along with all its contents. After navigating into the **dir2** directory and listing its contents, the files and subdirectories from **dir1** are visible.

```
satyajeet@vit: ~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file1      test
Documents Music      Public    Videos    dir1    file.txt  file2      thinclient_drives
satyajeet@vit:~$ mv test test_2026
satyajeet@vit:~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file1      test_2026
Documents Music      Public    Videos    dir1    file.txt  file2      thinclient_drives
satyajeet@vit:~$ |
```

12. mv old_name new_name – Moves or renames a file or directory.

Scenario 1 (Rename): As shown here, the command was used to rename the directory **test** to **test_2026** within the same location. After executing the command and listing the contents, the directory appears with the new name **test_2026**, confirming that the rename operation was successful.

```
satyajeet@vit: ~/test_2026$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file1      test_2026
Documents Music      Public    Videos    dir1    file.txt  file2      thinclient_drives
satyajeet@vit:~$ mv file1 test_2026
satyajeet@vit:~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file2      thinclient_drives
Documents Music      Public    Videos    dir1    file.txt  test_2026
satyajeet@vit:~$ cd test_2026/
satyajeet@vit:~/test_2026$ ls
file1
satyajeet@vit:~/test_2026$ |
```

Scenario 2 (Move): As shown here, the command was used to move the file **file1** into the directory **test_2026**. After executing the command, **file1** no longer appears in the home directory. Upon navigating into the **test_2026** directory and listing its contents, **file1** is visible there, confirming that the move operation was successful.

```
satyajeet@vit: ~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      file2      thinclient_drives
Documents Music      Public    Videos    dir1    file.txt  test_2026
satyajeet@vit:~$ rm file.txt
satyajeet@vit:~$ ls
Desktop  Downloads  Pictures  Templates  a      dir2      test_2026
Documents Music      Public    Videos    dir1    file2      thinclient_drives
satyajeet@vit:~$ |
```

13. rm file_name – Deletes a file permanently from the current directory.

As shown here, the file **file.txt** was removed using the **rm** command. After executing the command and listing the contents of the directory again, **file.txt** no longer appears, confirming that the file was successfully deleted.

```
satyajeet@vit: ~  
satyajeet@vit:~$ ls  
Desktop Downloads Pictures Templates a dir2 lost_data thinclient_drives  
Documents Music Public Videos dir1 file2 test_2026  
satyajeet@vit:~$ rm -r lost_data  
satyajeet@vit:~$ ls  
Desktop Downloads Pictures Templates a dir2 test_2026  
Documents Music Public Videos dir1 file2 thinclient_drives  
satyajeet@vit:~$ |
```

14. rm -r folder_name – Deletes a directory and all its contents recursively.

As shown here, the directory **lost_data** was removed using the recursive delete option. After executing the command and listing the contents of the directory again, **lost_data** no longer appears, confirming that the directory and all files inside it were permanently deleted.

[View & Read Files](#)

```
satyajeet@vit: ~  
satyajeet@vit:~$ ls  
Desktop Downloads Pictures Templates a dir2 linux.txt thinclient_drives  
Documents Music Public Videos dir1 file2 test_2026  
satyajeet@vit:~$ cat linux.txt  
Linux is a free, open-source operating system (OS) kernel, created by Linus Torvalds in 1991,  
that forms the foundation for many operating systems (like Ubuntu, Fedora, Android) used on  
everything from smartphones and PCs to the world's fastest supercomputers, known for its stab  
ility, security, and flexibility, powering much of the internet's servers and devices.  
satyajeet@vit:~$ |
```

15. cat file_name – Displays the contents of a file on the terminal.

As shown here, the contents of the file **linux.txt** are displayed using the **cat** command. The text stored inside **linux.txt** appears directly on the terminal, confirming that the file contains readable data and that the command successfully outputs the file's contents.

```
satyajee@vit: ~  
European colony in what is now the continental United States. British colonization followed  
with the 1607 settlement of Virginia, the first of the Thirteen Colonies. Enslavement of Afri  
cans was practiced in all colonies by 1770 and supplied most of the labor for the Southern Co  
lonies' plantation economy. Clashes with the British Crown began as a civil protest over the  
illegality of taxation without representation in Parliament and the denial of other English r  
ights. They evolved into the American Revolution, which led to the Declaration of Independenc  
e and a society based on universal rights. Victory in the 1775-1783 Revolutionary War brought  
international recognition of U.S. sovereignty and fueled westward expansion, further dispos  
sessing native inhabitants. As more states were admitted, a North-South division over slavery  
led the Confederate States of America to declare secession and fight the Union in the 1861-18  
65 American Civil War. With the United States' victory and reunification, slavery was abolish  
ed nationally. By 1900, the country had established itself as a great power, a status solidif  
ied after its involvement in World War I. Following Japan's attack on Pearl Harbor in 1941, t  
he U.S. entered World War II. Its aftermath left the U.S. and the Soviet Union as rival super  
powers, competing for ideological dominance and international influence during the Cold War.  
The Soviet Union's collapse in 1991 ended the Cold War, leaving the U.S. as the world's sole  
superpower.  
  
The U.S. national government is a presidential constitutional federal republic and representa  
tive democracy with three separate branches: legislative, executive, and judicial. It has a b  
icameral national legislature composed of the House of Representatives (a lower house based o  
n population) and the Senate (an upper house based on equal representation for each state). F  
ederalism grants substantial autonomy to the 50 states. In addition, 574 Native American trib  
es have sovereignty rights, and there are 326 Native American reservations. Since the 1850s,  
the Democratic and Republican parties have dominated American politics. American ideals and v  
alues are based on a democratic tradition inspired by the American Enlightenment movement.  
(END)
```

16. less file_name – Displays the contents of a file one screen at a time, allowing easy navigation.

As shown here, the **less** command is used to open a large text file in an interactive mode. The content is displayed one screen at a time, allowing smooth navigation using the keyboard. The **(END)** indicator at the bottom shows that the end of the file has been reached, confirming that the file content was successfully viewed using **less**.

```
satyajeet@vit:~$ cat data.txt
Line 1 - This is sample data for line one
Line 2 - This is sample data for line two
Line 3 - This is sample data for line three
Line 4 - This is sample data for line four
Line 5 - This is sample data for line five
Line 6 - This is sample data for line six
Line 7 - This is sample data for line seven
Line 8 - This is sample data for line eight
Line 9 - This is sample data for line nine
Line 10 - This is sample data for line ten
Line 11 - This is sample data for line eleven
Line 12 - This is sample data for line twelve
Line 13 - This is sample data for line thirteen
Line 14 - This is sample data for line fourteen
Line 15 - This is sample data for line fifteen
```

```
satyajeet@vit:~$ head data.txt
Line 1 - This is sample data for line one
Line 2 - This is sample data for line two
Line 3 - This is sample data for line three
Line 4 - This is sample data for line four
Line 5 - This is sample data for line five
Line 6 - This is sample data for line six
Line 7 - This is sample data for line seven
Line 8 - This is sample data for line eight
Line 9 - This is sample data for line nine
Line 10 - This is sample data for line ten
satyajeet@vit:~$ |
```

17. head file_name – Displays the first few lines of a file (by default, the first 10 lines).

As shown here, the file `data.txt` contains 15 lines when viewed using the `cat` command. When the `head data.txt` command is executed, only the first 10 lines (Line 1 to Line 10) are displayed on the terminal, while the remaining lines (Line 11 to Line 15) are not shown, confirming the default behavior of the `head` command.


```
satyajeeet@vit:~$ tail data.txt
Line 6 - This is sample data for line six
Line 7 - This is sample data for line seven
Line 8 - This is sample data for line eight
Line 9 - This is sample data for line nine
Line 10 - This is sample data for line ten
Line 11 - This is sample data for line eleven
Line 12 - This is sample data for line twelve
Line 13 - This is sample data for line thirteen
Line 14 - This is sample data for line fourteen
Line 15 - This is sample data for line fifteen
satyajeeet@vit:~$ |
```

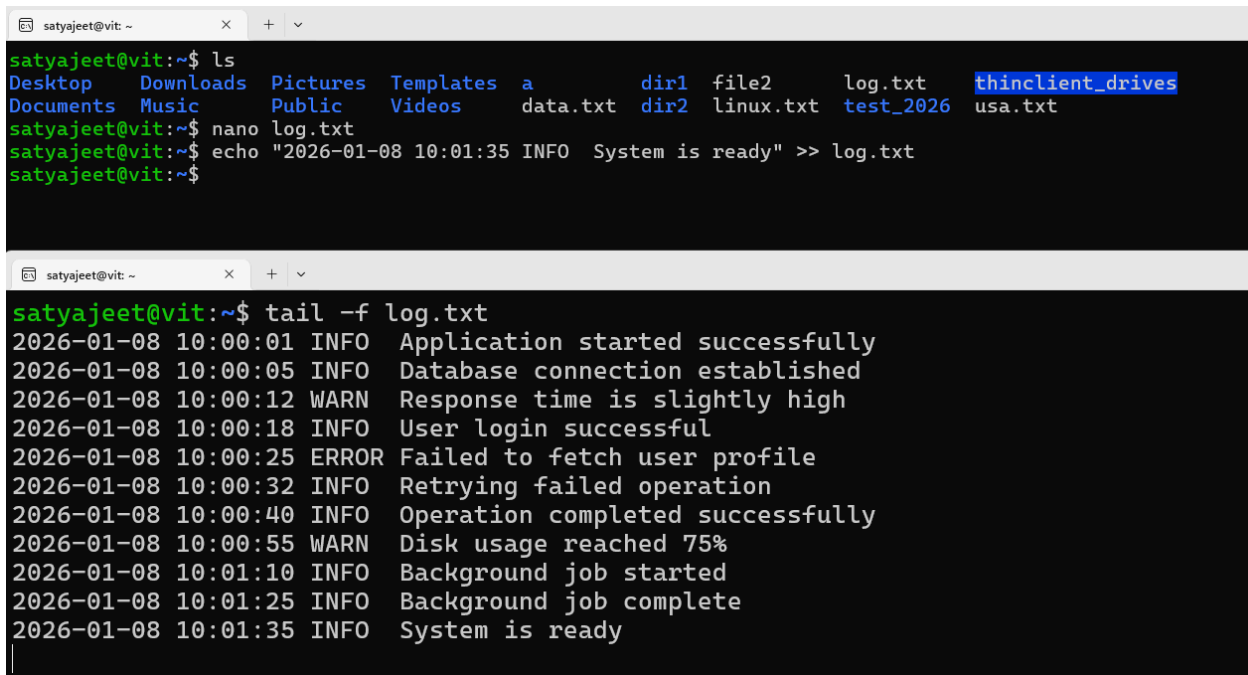
18. tail file_name – Displays the last 10 lines of a file by default.

As shown here, when the `tail data.txt` command is executed on the file **data.txt**, only the last 10 lines are displayed on the terminal. Since the file contains 15 lines in total, the output starts from **Line 6** and goes up to **Line 15**, confirming that the `tail` command shows the ending portion of a file by default.

```
satyajeeet@vit:~$ tail -f log.txt
2026-01-08 10:00:01 INFO Application started successfully
2026-01-08 10:00:05 INFO Database connection established
2026-01-08 10:00:12 WARN Response time is slightly high
2026-01-08 10:00:18 INFO User login successful
2026-01-08 10:00:25 ERROR Failed to fetch user profile
2026-01-08 10:00:32 INFO Retrying failed operation
2026-01-08 10:00:40 INFO Operation completed successfully
2026-01-08 10:00:55 WARN Disk usage reached 75%
2026-01-08 10:01:10 INFO Background job started
2026-01-08 10:01:25 INFO Background job complete
2026-01-08 10:01:35 INFO System is ready
```

19. tail -f file_name – Displays the last lines of a file and continuously monitors it for new entries in real time.

As shown here, the **tail -f log.txt** command is used to view live logs from the file **log.txt**. The existing log entries are displayed first, and the command keeps running, waiting for new log messages to be appended. Any new entries added to **log.txt** will appear instantly on the terminal, confirming real-time log monitoring.



The screenshot shows a terminal window with the following commands and output:

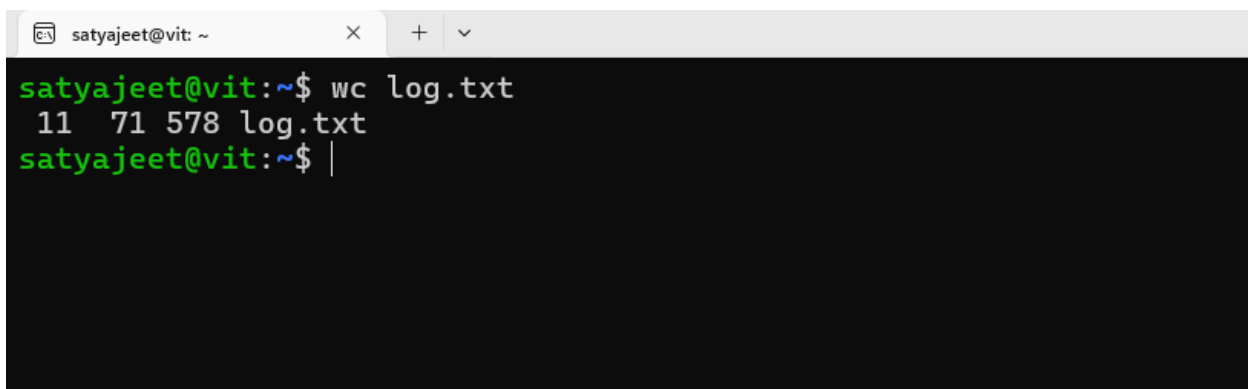
```
satyajeet@vit: ~$ ls
Desktop  Downloads  Pictures  Templates  a      dir1  file2  log.txt  thinclient_drives
Documents Music      Public    Videos    data.txt dir2  linux.txt test_2026 usa.txt

satyajeet@vit: ~$ nano log.txt
satyajeet@vit: ~$ echo "2026-01-08 10:01:35 INFO System is ready" >> log.txt
satyajeet@vit: ~$
```

Below this, the **tail -f log.txt** command is executed, displaying the following log entries:

```
2026-01-08 10:00:01 INFO Application started successfully
2026-01-08 10:00:05 INFO Database connection established
2026-01-08 10:00:12 WARN Response time is slightly high
2026-01-08 10:00:18 INFO User login successful
2026-01-08 10:00:25 ERROR Failed to fetch user profile
2026-01-08 10:00:32 INFO Retrying failed operation
2026-01-08 10:00:40 INFO Operation completed successfully
2026-01-08 10:00:55 WARN Disk usage reached 75%
2026-01-08 10:01:10 INFO Background job started
2026-01-08 10:01:25 INFO Background job complete
2026-01-08 10:01:35 INFO System is ready
```

As shown here, a new terminal window was opened and a new log entry was appended to the file **log.txt** using the **echo** command. While **tail -f log.txt** was already running in another terminal, the newly added log entry (**System is ready**) appeared immediately in the live log output, confirming that **tail -f** continuously monitors the file and displays new entries in real time.



The screenshot shows a terminal window with the following command and output:

```
satyajeet@vit: ~$ wc log.txt
11  71 578 log.txt
satyajeet@vit: ~$
```

20. wc file_name – Displays the count of lines, words, and characters in a file.

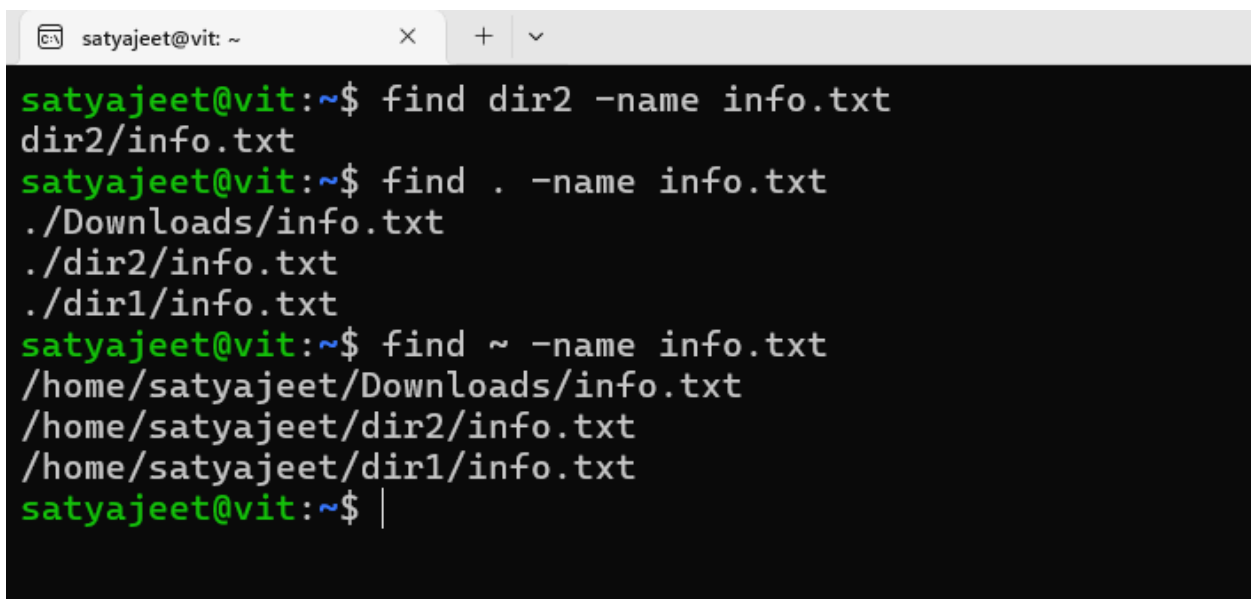
As shown here, when the `wc log.txt` command is executed, the output shows three values:

the first number (11) represents the total number of lines,

the second number (71) indicates the total number of words,

and the third number (578) shows the total number of characters present in the file `log.txt`.

[Search & Filter](#)



```
satyajeet@vit: ~  
satyajeet@vit:~$ find dir2 -name info.txt  
dir2/info.txt  
satyajeet@vit:~$ find . -name info.txt  
./Downloads/info.txt  
./dir2/info.txt  
./dir1/info.txt  
satyajeet@vit:~$ find ~ -name info.txt  
/home/satyajeet/Downloads/info.txt  
/home/satyajeet/dir2/info.txt  
/home/satyajeet/dir1/info.txt  
satyajeet@vit:~$ |
```

21. find path -name file_name – Searches for a file with a specific name within a given directory and all its subdirectories.

As shown here, the command `find dir2 -name info.txt` searches only inside the `dir2` directory and returns `dir2/info.txt`. When `find . -name info.txt` is executed, the search starts from the current directory and lists all matching files under it, including `./Downloads/info.txt`, `./dir2/info.txt`, and `./dir1/info.txt`. When `find ~ -name info.txt` is used, the search begins from the home directory (`/home/satyajeet`) and displays the full paths of all matching files found within the home directory structure.

```
satyajeet@vit: ~$ grep "is" usa.txt
The United States of America (USA), also known as the United States (U.S.) or America, is a country primarily located in North America. It is a federal republic of 50 states and a federal capital district, Washington, D.C. The 48 contiguous states border Canada to the north and Mexico to the south, with the semi-exclave of Alaska in the northwest and the archipelago of Hawaii in the Pacific Ocean. The United States also asserts sovereignty over five major island territories and various uninhabited islands in Oceania and the Caribbean.[j] It is a megadiverse country, with the world's third-largest land area[c] and third-largest population, exceeding 340 million.[k]
```

22. grep "pattern" file_name – Searches for lines in a file that contain a specified text pattern.

As shown here, the command **grep "is" usa.txt** searches for the word **is** inside the file **usa.txt**. All lines containing the matching text are displayed on the terminal, and each occurrence of the searched pattern is highlighted, making it easier to identify where the match appears within the file.

```
satyajeet@vit: ~$ grep -r "info" /home/satyajeet/dir1
/home/satyajeet/dir1/info.txt:This is information file.
/home/satyajeet/dir1/data.txt:So for you kind information, don't delete it.
satyajeet@vit: ~$
```

23. grep -r "pattern" folder_path – Recursively searches for a specified text pattern inside all files within a directory and its subdirectories.

As shown here, the command **grep -r "info" /home/satyajeet/dir1** searches for the text **info** inside every file present in the directory **/home/satyajeet/dir1**. The output displays the full file paths (**info.txt** and **data.txt**) along with the matching lines, and the searched text is highlighted, confirming that the pattern was found in multiple files within the directory.

```
satyajeet@vit: ~  
satyajeet@vit:~$ sudo apt install plocate  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
plocate is already the newest version (1.1.19-2ubuntu2).  
The following package was automatically installed and is no longer required:  
  libllvm19  
Use 'sudo apt autoremove' to remove it.  
0 upgraded, 0 newly installed, 0 to remove and 55 not upgraded.  
satyajeet@vit:~$ sudo updatedb  
satyajeet@vit:~$  
satyajeet@vit:~$ |
```

As shown here, before using the **locate** command, the **plocate** package must be installed and the file index database must be updated using **sudo updatedb**. This step ensures that the database contains the latest file and directory information, allowing the **locate** command to return accurate and instant search results.

```
satyajeet@vit: ~  
satyajeet@vit:~$ locate info.txt  
/home/satyajeet/Downloads/info.txt  
/home/satyajeet/dir1/info.txt  
/home/satyajeet/dir2/info.txt  
/home/satyajeet/test_2026/info.txt  
/mnt/c/Program Files/Microsoft Office/root/Office16/Configuration/ssn_high_group_info.txt  
/mnt/c/Program Files (x86)/Yawcam/logs/info.txt  
/mnt/c/Users/Satyajeet/.yawcam/logs/info.txt  
satyajeet@vit:~$ |
```

24. locate file_name – Quickly searches for files by name using a prebuilt system database.

As shown here, when the **locate info.txt** command is executed, it instantly displays all paths where files named **info.txt** exist across the system. The results include locations inside the home directory such as **Downloads**, **dir1**, **dir2**, and **test_2026**, as well as files found in other mounted paths. This confirms that **locate** searches the indexed database rather than scanning directories in real time, making it much faster than the **find** command.

Permissions & Users

```
satyajeet@vit: ~  
satyajeet@vit:~$ ls  
Desktop    Music      Templates  data.txt  file2      test_2026  
Documents  Pictures   Videos    dir1      linux.txt  thinclient_drives  
Downloads  Public     a          dir2      log.txt    usa.txt  
satyajeet@vit:~$ ls -l log.txt  
-rw-r--r-- 1 satyajeet satyajeet 578 Jan  7 20:38 log.txt  
satyajeet@vit:~$ chmod 755 log.txt  
satyajeet@vit:~$ ls -l log.txt  
-rwxr-xr-x 1 satyajeet satyajeet 578 Jan  7 20:38 log.txt  
satyajeet@vit:~$ |
```

25. chmod 755 file_name – Changes file permissions using numeric (octal) values.

As shown here, before running the command, the file `log.txt` had permissions `-rw-r--r--`, meaning the owner could read and write, while the group and others could only read. After executing `chmod 755 log.txt`, the permissions changed to `-rwxr-xr-x`, making the file executable.

Permission calculation (numeric method):

- 4 → Read (r)
- 2 → Write (w)
- 1 → Execute (x)

The number 755 is calculated as follows:

- 7 (owner) = 4 (read) + 2 (write) + 1 (execute) → rwx
- 5 (group) = 4 (read) + 1 (execute) → r-x
- 5 (others) = 4 (read) + 1 (execute) → r-x

As a result, the owner gets full permissions, while the group and others can read and execute the file but cannot modify it.

```
satyajeet@vit: ~  
satyajeet@vit:~$ sudo groupadd developers  
satyajeet@vit:~$ sudo chown :developers log.txt  
satyajeet@vit:~$ ls -l log.txt  
-rwxr-xr-x 1 satyajeet developers 578 Jan  7 20:38 log.txt  
satyajeet@vit:~$ sudo chown :satyajeet log.txt  
satyajeet@vit:~$ ls -l log.txt  
-rwxr-xr-x 1 satyajeet satyajeet 578 Jan  7 20:38 log.txt  
satyajeet@vit:~$ |
```

26. sudo chown :group_name file_name – Changes only the group ownership of a file without affecting the owner.

As shown here, a new group developers was first created using the groupadd command. The group ownership of log.txt was then changed from satyajeet to developers using **sudo chown :developers log.txt**. After verification with **ls -l**, the group ownership was reverted back to satyajeet using **sudo chown :satyajeet log.txt**, confirming that the group of a file can be modified and restored independently of the file owner.

```
satyajeet@vit: ~  
satyajeet@vit:~$ ls -l  
total 72  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Desktop  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Documents  
drwxr-xr-x 2 satyajeet satyajeet 4096 Jan 7 21:26 Downloads  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Music  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Pictures  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Public  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Templates  
drwxr-xr-x 2 satyajeet satyajeet 4096 Dec 9 20:33 Videos  
drwxr-xr-x 3 satyajeet satyajeet 4096 Jan 7 13:50 a  
-rw-r--r-- 1 satyajeet satyajeet 674 Jan 7 20:28 data.txt  
drwxr-xr-x 3 satyajeet satyajeet 4096 Jan 7 21:48 dir1  
drwxr-xr-x 3 satyajeet satyajeet 4096 Jan 7 19:04 dir2  
-rw-r--r-- 1 satyajeet satyajeet 20 Jan 7 18:46 file2  
-rw-r--r-- 1 satyajeet satyajeet 366 Jan 7 20:13 linux.txt  
-rwxr-xr-x 1 satyajeet satyajeet 578 Jan 7 20:38 log.txt  
drwxr-xr-x 2 satyajeet satyajeet 4096 Jan 7 21:44 test_2026  
drwxrwxr-t 2 satyajeet satyajeet 4096 Dec 9 20:33 thinclient_drives  
-rw-r--r-- 1 satyajeet satyajeet 3109 Jan 7 20:18 usa.txt  
satyajeet@vit:~$ |
```

27. ls -l – Displays files and directories in long listing format with detailed information.

As shown here, the `ls -l` command lists each file and directory along with its permissions, number of links, owner, group, size, last modified date, and name. Entries starting with `d` represent directories, while `-` indicates regular files. The permission string (such as `-rwxr-xr-x`) shows access rights for the owner, group, and others. Additionally, the directory `thinclient_drives` displays a `t` (sticky bit) at the end of its permissions, indicating that only the file owner or root can delete files within that directory, even though it is writable by others.

```
satyajeet@vit: ~  
satyajeet@vit:~$ whoami  
satyajeet  
satyajeet@vit:~$ |
```

28. whoami – Displays the username of the currently logged-in user.

As shown here, when the `whoami` command is executed, it returns `satyajeet`, confirming the identity of the user under which the current terminal session is running. This command is commonly used to verify the active user, especially while working with permissions, ownership, or switching users.

```
satyajeet@vit: ~$ apt update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
satyajeet@vit: ~$ sudo apt update
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1685 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [1393 kB]
```

29. sudo command – Executes a command with administrative (root) privileges.

As shown here, running `apt update` without `sudo` fails due to permission restrictions on system directories such as `/var/lib/apt` and `/var/cache/apt`. When the same command is executed using `sudo`, the system temporarily grants administrative privileges, allowing the package index to be updated successfully from the configured Ubuntu repositories. This demonstrates that package management operations require elevated permissions.

```
satyajeet@vit: ~$
satyajeet@vit: ~$
satyajeet@vit: ~$
satyajeet@vit: ~$ id
uid=1000(satyajeet) gid=1000(satyajeet) groups=1000(satyajeet),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
satyajeet@vit: ~$
```

30. id – Displays the user ID (UID), primary group ID (GID), and all groups the current user belongs to.

As shown here, when the `id` command is executed, it displays the UID and GID for the user **satyajeet**, along with the list of groups such as `satyajeet`, `adm`, `cdrom`, `sudo`, and others. This confirms the user's identity and group memberships, which is especially useful while working with permissions, ownership, and administrative access.